



PES University, Bengaluru
(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - DEC 2025

UE23MA241B - Linear Algebra and its Applications

Total Marks : 100.0

1.a. When does the breakdown of Gaussian elimination occur? How can this issue be addressed? (7.0 Marks)

1.b. Use the Gauss-Jordan method to find the inverse of the matrix (10.0 Marks)

$$A = \begin{bmatrix} 1 & -2 & -2 \\ -1 & 3 & 2 \\ -1 & 1 & 1 \end{bmatrix}.$$

1.c. Find the matrices L and U such that $A=LU$ for the given matrix. (8.0 Marks)

$$A = \begin{bmatrix} 2 & -1 & 1 \\ 3 & 3 & 9 \\ 3 & 3 & 5 \end{bmatrix}$$

2.a. Find the four fundamental subspaces-the row space, column space, null space, and left null space along with their basis, dimension, and the rank of the given 3×4 matrix:

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix}.$$

(12.0 Marks)

2.b. Define the following:

i) Linear Independence ii) Basis and Dimension iv) Rank v) Span

(8.0 Marks)

2.c. Determine whether the following set of vectors is linearly independent or not.

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \quad \vec{v}_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$$

(5.0 Marks)

3.a. Apply the following transformations on the given matrix A and find the resultant matrices.

$$\begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$$

- i) Apply scaling with a scaling factor $s_x=4$ along the x-axis and $s_y=0.5$ along the y-axis.
- ii) Apply rotation in the counter clockwise direction by 60° .
- iii) Reflect the matrix along the line $y=-x$.
- iv) Project the matrix orthogonally onto the x-axis. (10.0 Marks)

3.b. . Find the least squares solution for the given system of equations:

$$3x+4y=10$$

$$2x+y=5$$

(10.0 Marks)

3.c. Mention any four properties of eigenvalues and eigenvectors

(5.0 Marks)

4.a. Find SVD of the given matrix

(12.0 Marks)

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 4 & 0 & 6 \end{bmatrix}$$

4.b. Explain any three applications of SVD.

(6.0 Marks)

4.c. Compute the covariance matrix of the dataset, where each row represents an observation and each column represents a variable.

(7.0 Marks)

$$X = \begin{bmatrix} 2 & 4 \\ 4 & 6 \\ 6 & 8 \end{bmatrix}$$

END SEMESTER ASSESSMENT (ESA) - JULY - 2023**UE21MA241B - Linear Algebra****Total Marks : 100.0**

1.a.

Explain the column approach to solve the system of equations $x - 2y = 3$ and $2x + y = 1$ with a neat diagram.

(6.0 Marks)

1.b.

Do the three planes $x + 2y + z = 4$, $y - z = 1$, $x + 3y = 0$ have at least one common point of the intersection? Explain. Is the system consistent if the last is changed to $x + 3y = 5$. If so, solve the system completely.

(7.0 Marks)

1.c.

Use Gauss Jordan method to obtain the inverse of $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$.

(7.0 Marks)

(4.0 Marks)

2.a. Check whether $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix}$ are independent in R^3 .

2.b.

Find the bases and dimension of the four fundamental subspaces of the matrix

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 3 \\ 3 & 6 & 3 & 7 \end{bmatrix}.$$

(11.0 Marks)

2.c.

Reduce $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ to echelon form and find its rank. Identify the pivot and free variables.

(5.0 Marks)

3.a.

On the space P_3 of cubic polynomials, what matrix represents $\frac{d^2}{dt^2}$? Construct 4 by 4 matrix from the standard basis $1, t, t^2, t^3$. Find its column space and null space.

(7.0 Marks)

3.b.

Find all vectors in R^3 that are orthogonal to $(1, 1, 1)$ and $(1, -1, 0)$. Produce an orthonormal basis from these vectors (mutually orthogonal unit vectors).

(6.0 Marks)

3.c.

Find the projection of b onto the column space of A , where $A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ -2 & 4 \end{bmatrix}$ and

$$b = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

(7.0 Marks)

4.a.

Find an orthonormal set q_1, q_2, q_3 for which q_1, q_2 span the column space of

$A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix}$. Which fundamental subspace contain q_3 . What is the least square solution of $Ax = b$ if $b = [1 \ 2 \ 7]^T$.

(7.0 Marks)

4.b.

Find the eigen values and corresponding eigen vectors of $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$.

(7.0 Marks)

4.c.

Diagonalize the matrix $A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$ and find one of its square roots, a matrix R such that $R^2 = A$. How many such square root matrices are there?

(6.0 Marks)

5.a.

Check the positive definiteness (any two methods) of $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & 2 \\ 2 & 2 & 3 \end{bmatrix}$.

(5.0 Marks)

5.b.

Write the symmetric matrix which corresponds to the following quadratic form,
 $Q(x) = x_1^2 + x_2^2 + x_3^2 - 2x_1x_2 - 2x_1x_3 + 2x_2x_3$.

(4.0 Marks)

(11.0 Marks)

5.c. Find SVD of the matrix $A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$.

END SEMESTER ASSESSMENT (ESA) - JULY - 2023**UE20MA251 - Linear Algebra and Its Applications****Total Marks : 100.0**

1.a.

Solve the following system of equations using Gaussian elimination:

(6.0 Marks)

$$2x + y + 3z = 1, 2x + 6y + 8z = 3, 6x + 8y + 18z = 5$$

1.b. Write down the elementary matrices E, F, G associated with the system of equations $2u + v + 3w = -1$, $4u + v + 7w = 5$, $-6u - 2v - 12w = -2$. Also find the LU decomposition of A.

(7.0 Marks)

Use the Gauss – Jordan method to find the inverse of:

(7.0 Marks)

1.c.

$$A = \begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}.$$

Find the value(s) of 'h' for which the following set of vector's,

(6.0 Marks)

2.a. $\left\{ \mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} h \\ 1 \\ -h \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 1 \\ 2h \\ 3h+1 \end{bmatrix} \right\}$

Is linearly independent.

- 2.b. Find a basis and the dimension of the subspaces $V = \{(a, b, 0) : a \text{ and } b \text{ are real numbers}\}$, $W = \{(0, b, c) : b \text{ and } c \text{ are real numbers}\}$ and $V \cap W$ in R^3 . Geometrically describe the basis of V, W and $V \cap W$.

(7.0 Marks)

If $A = \begin{pmatrix} 0 & 1 & 3 \\ -1 & 0 & 1 \\ -1 & 2 & 7 \end{pmatrix}$,

(7.0 Marks)

- 2.c. (i) Find the basis for the column space of A.
(ii) Find the basis for the Row space of A.
(iii) Find the basis for the Null space of A.
(iv) Is the vector $w = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ is in Row space of A. (3+1+2+1=7M)

- 3.a. Let $T: R^3 \rightarrow R^3$ given by $T \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2x + y + z \\ x - y + 2z \\ x + 2y - z \end{bmatrix}$, then find Null space of T i.e., $N(T)$.

(6.0 Marks)

- 3.b. What matrix P projects every point in $\mathbb{R}^3$ onto the line of intersection of the planes $x + y + z = 0$ and $x - z = 0$? What are the column space and row space of matrix P .

(7.0 Marks)

3.c.

Consider the linear system $-x_1 + x_2 = 10$, $2x_1 + x_2 = 5$, $x_1 - 2x_2 = 20$. This system is over determined and inconsistent. Find the least approximation to b and the least squares solution $\hat{x}$ to this system and rewrite the linear system as $A\hat{x} = b_1$

(7.0 Marks)

- 4.a. Find a third column so that the matrix $Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{14}} & - & - & - \\ \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{14}} & - & - & - \\ \frac{1}{\sqrt{3}} & \frac{-3}{\sqrt{14}} & - & - & - \end{bmatrix}$ is orthogonal.

(5.0 Marks)

- 4.b. Find the Eigen Value and Eigen vectors of $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$

(7.0 Marks)

- 4.c. Use Gram Schmidt process to find the orthonormal vectors q_1, q_2, q_3 form the given independent vectors $a = (1, 1, 1), b = (0, 1, 1), c = (0, 0, 1)$, and also factorize $A=QR$ (8.0 Marks)

- 5.a. For the semi definite matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$, write $X^T A X$ as the sum of two squares. (6.0 Marks)

- 5.b. Let $Q(x) = x_1^2 + bx_2^2 + 7x_3^2 + 4x_1x_2 + 16x_2x_3 + 8x_1x_3$, for what range of numbers b , $Q(x)$ is positive definite. (4.0 Marks)

- 5.c. Find the SVD of $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}_{2 \times 3}$ (10.0 Marks)



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UE19MA251**JULY 2021: END SEMESTER ASSESSMENT (ESA) B TECH IV SEMESTER****UE19MA251 - Linear Algebra and Its Applications**

Time: 3 Hrs

Answer All Questions

Max Marks: 100

1	a)	Solve the system of equations if possible using Gaussian Elimination: $x+y+2z+3t=13$, $x-2y+z+t=8$, $3x+y+z-t=1$	7
	b)	Write down the elementary matrices E,F,G associated with the system of equations $2u+3v-4w=3$, $u+2v+2w=10$. $u+v+w=7$. Also find A=LU factorization for A if possible.	6
	c)	Use Gauss-Jordan method to find A^{-1} if it exists for $A = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{pmatrix}$	7
2	a)	If $A = \begin{pmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{pmatrix}$ find the basis and dimension of Nullspace and Left Nullspace	7

		of A	
	b)	Examine if the set of vectors $\{(1,2,-1), (1,-2,1), (3,-2,1)\}$ is linearly dependent/independent. If the set is dependent find a relation between the vectors	6
	c)	Find three independent vectors on the plane $P: x+2y-3z-t=0$ in R^4 . This plane is the nullspace of what matrix? Find a basis for the orthogonal complement of P?	7
3	a)	If V is a subspace of R^5 spanned by $(1,2,3,-1,2)$ and $(2,4,7,2,-1)$. Find a basis of the orthogonal complement $V^\perp$ of V .	7
	b)	Suppose a linear transformation T transforms $(1,1)$ to $(2,2)$ and $(2,0)$ to $(0,0)$. Find $T(v)$ when (a) $v=(2,2)$, (b) $v=(3,1)$ (c) $v=(-1,1)$	6

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		$A = \begin{pmatrix} 1 & 3 & 2 \\ -1 & -4 & -3 \end{pmatrix}$ using the Cayley – Hamilton theorem.	
	c)	Diagonalize the matrix $A = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ and find one of its square roots (Square root of A is a matrix R such that $R^2 = A$)	7
5	a)	For which s and t do A and B have all $\lambda > 0$ (or pivots > 0) and are therefore positive definite. $A = \begin{pmatrix} s & -4 & -4 \\ -4 & s & -4 \\ -4 & -4 & s \end{pmatrix}$ $B = \begin{pmatrix} 1 & 2 & 4 \\ 2 & t & 8 \\ 4 & 8 & 7 \end{pmatrix}$	6
	b)	If $A = LDL^T = L\sqrt{D}\sqrt{D}L^T = R^T R$ is symmetric positive definite, then find $R = \sqrt{D}L^T$ for $A = \begin{pmatrix} 4 & 8 \\ 8 & 25 \end{pmatrix}$	6
	c)	Find SVD of the following matrices: $\begin{pmatrix} 4 & 4 \\ -3 & 3 \end{pmatrix}$	8

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END SEMESTER ASSESSMENT (ESA) - JULY - 2023**UE19MA251 - Linear Algebra and Its Applications****Total Marks : 100.0**

1.a. Determine the values of a and b for which the system of equations
 $x + y + az = 2b$,
 $x + 3y + (2 + 2a)z = 7b$
 $3x + y + (3 + 3a)z = 11b$ Will have (i) unique nontrivial solution (ii) trivial solution (iii) no solution (iv) infinity of solutions. (5.0 Marks)

1.b. A boy is walking along the path $y = ax^2 + bx + c$ through the points $(-6, 8)$, $(2, 12)$, and $(3, 8)$.
He wants to meet his friend at $P(7, 60)$.
Will he meet his friend? (Use Gaussian elimination method.) (5.0 Marks)

1.c. Solve the system of equation by using LU Decomposition.
 $2x + 3y + z = 9$,
 $x + 2y + 3z = 6$,
 $3x + y + 2z = 8$. (5.0 Marks)

1.d. . Solve the following system by using the Gauss-Jordan elimination method.

$$2x + y + z = 0$$

$$4x - 6y = 0$$

$$-2x + 7y + 2z = 0$$

(5.0 Marks)

2.a. Examine if the following set of vectors are linearly independent. When the set is dependent find a relation between the vectors:

$$\{(1, 3, 1, 2), (2, 5, -1, 2), (1, 3, 7, -2)\}$$

(5.0 Marks)

2.b. Reduce the following matrix to Row Reduced Echelon form and determine their ranks,

Identify the pivot variables and free variables. Find the special solutions to $Ax=0$.

(5.0 Marks)

$$\begin{pmatrix} 2 & -4 & 4 & -2 \\ 4 & -9 & 7 & -3 \\ 1 & -4 & 8 & 0 \end{pmatrix}$$

2.c. Find the Column space and Null space for the following matrices:

(5.0 Marks)

$$\begin{pmatrix} 2 & 4 & -2 & 2 \\ -2 & 5 & 7 & 3 \\ -3 & 6 & -8 & 6 \end{pmatrix}$$

2.d. For which vector (b_1, b_2, b_3, b_4) is this system solvable?

(5.0 Marks)

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 5 & 7 \\ 3 & 9 & 12 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix}$$

3.a. Find the matrix of the linear transformation T on $\mathbb{R}^3$ defined by $T(x,y,z)=(x+2y-z, y+z, x+y-2z)$ with respect to the non-standard basis $(1,1,1), (1,1,0), ((1,0,0)$

(5.0 Marks)

3.b. Project $b = (1, 0, 0)$ onto the lines through $a_1 = (-1, 2, 2)$, $a_2 = (2, 2, -1)$ and $a_3 = (2, -1, 2)$. Add the three projections to get the sum b . Also find the corresponding projection matrices P_1, P_2 and P_3 . Check that their sum is I and the product is 0 .

(5.0 Marks)

3.c. Find the point on the plane $2x+5y+z=0$ that is nearest to $(2,0,1)$

(5.0 Marks)

3.d. An ice-cream vendor records the number of hours of sun shine (x) versus the number of ice-creams sold in an hour (y) at his shop from Monday to Friday and found the following data :

x: 2 3 5 7 9

y: 4 5 7 10 15

Find the best values of m and c that suit the equation $y = mx + c$. If there is a weather forecast that says there would be 8 hours of sun shine the next day, estimate the number of ice-creams that he expects to sell on that day. (5.0 Marks)

4.a. If W is a subspace spanned by the orthogonal vectors (2, 5, -1) and (-2, 1, 1) find the point in W that is closest to (1, 2, 3). (5.0 Marks)

4.b. Find the eigenvalues and the corresponding eigenvectors of $\begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix}$. (5.0 Marks)

4.c.

Find the matrix A whose eigen values are 2 and 5, and whose eigen vectors are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ respectively using $S\Lambda S^{-1}$.

(5.0 Marks)

4.d. Find the matrices S and S^{-1} to diagonalize $A = \begin{pmatrix} 0.6 & 0.4 \\ 0.4 & 0.6 \end{pmatrix}$

What are limits of Δ^k and $S\Delta^k S^{-1}$ as $k \rightarrow \infty$?

(5.0 Marks)

5.a. For which a are the matrix A have all

$\lambda > 0$ and are therefore positive definite. $A = \begin{pmatrix} a & 2 & 2 \\ 2 & a & 2 \\ 2 & 2 & a \end{pmatrix}$

(5.0 Marks)

5.b.

Compute the quadratic form $x^T A x$ for $A = \begin{pmatrix} 4 & 3 & 0 \\ 3 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ and

(5.0 Marks)

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

5.c. Find SVD of the following matrix $\begin{pmatrix} 4 & 4 \\ -3 & 3 \end{pmatrix}$

(5.0 Marks)

5.d. The following table lists the weights and heights of 5 boys:

Boy	1	2	3	4	5
Weight(lbs)	120	125	125	135	145
Height(in)	61	60	64	68	72

Find the covariance matrix for the above data

(5.0 Marks)

END SEMESTER ASSESSMENT (ESA) - JULY - 2023**UE17MA251 - Linear Algebra and Its Applications****Total Marks : 100.0**

1.a. Solve: $x + y + z = 6$, $2x - y + z = 3$, $x + z = 4$ by using Gauss elimination method.

(6.0 Marks)

(7.0 Marks)

1.b. Find LU factorization for $A = \begin{bmatrix} 2 & -3 & -1 & 2 & 3 \\ 4 & -4 & -1 & 4 & 11 \\ 2 & -5 & -2 & 2 & -1 \\ 0 & 2 & 1 & 0 & 4 \end{bmatrix}$

(7.0 Marks)

1.c. Find the inverse of $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$, using Gauss-Jordon method.

i) Define free and pivot variables.

- 2.a. ii) For every c , find R and special solutions to $Ax = 0$, where $A = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 1 & c & 2 & 2 \end{bmatrix}$
- (7.0 Marks)

2.b.

- i) Define basis and dimension for the vector space V with example.
- ii) If V is the subspace spanned by $(1,1,1)$ and $(2,1,0)$, find a matrix A that has V as its row space, find a matrix B that has V as its null space.

(6.0 Marks)

- 2.c. Find the four fundamental subspaces of the matrix $A = \begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ 3 & 1 & 0 & 1 \end{bmatrix}$
- (7.0 Marks)

3.a.

- I. Define linear transformation.
- II. Find the differentiation matrix A_{diff} which differentiate the fourth degree polynomials.

(7.0 Marks)

3.b.

Let $T_1: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the reflection about the line $y=x$ and $T_2: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the projection on x -axis. Find the standard matrix of $T_2 \cdot T_1$. Is this matrix same as the standard matrix of $T_1 \cdot T_2$? Illustrate your result graphically.

(7.0 Marks)

3.c.

Use the method least squares, find the equation of the line that runs through four points $(1,-1), (4,11), (-1,-9)$ and $(-2,-13)$.

(6.0 Marks)

4.a.

Find the matrices Q and R such that $QR = A$, where A has columns $(1,1,-2), (1,2,-3), (0,1,1)$.

(8.0 Marks)

4.b. Calculate the four iterations of the power method to find the largest Eigen value and corresponding Eigen vector of the matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$, use $(1, 1, 1)$ as initial approximation.

(5.0 Marks)

(7.0 Marks)

4.c. Diagonalize the matrix $A = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$ and hence find A^{50} .

Define positive definite and positive semi definite of the matrix. And test the following matrices for positive or positive semi definite

5.a. i) $\begin{bmatrix} 5 & 2 & 1 \\ 2 & 2 & 2 \\ 1 & 2 & 5 \end{bmatrix}$ ii) $\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$

(10.0 Marks)

5.b. Find Singular Value Decomposition (SVD) of the matrix $A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & 2 \end{bmatrix}$ (10.0 Marks)